

## 6.4 Thermal Information

| THERMAL METRIC <sup>(1)(2)</sup>                               | SN74LS00 |           |          |         | UNIT |
|----------------------------------------------------------------|----------|-----------|----------|---------|------|
|                                                                | D (SOIC) | DB (SSOP) | N (PDIP) | NS (SO) |      |
|                                                                | 14 PINS  | 14 PINS   | 14 PINS  | 14 PINS |      |
| $R_{\theta JA}$ Junction-to-ambient thermal resistance         | 90.9     | 102.8     | 54.8     | 89.7    | °C/W |
| $R_{\theta JC(top)}$ Junction-to-case (top) thermal resistance | 51.9     | 53.3      | 42.1     | 48.1    | °C/W |
| $R_{\theta JB}$ Junction-to-board thermal resistance           | 48       | 53.4      | 34.8     | 50.1    | °C/W |
| $\psi_{JT}$ Junction-to-top characterization parameter         | 18.6     | 16.5      | 26.9     | 16.7    | °C/W |
| $\psi_{JB}$ Junction-to-board characterization parameter       | 47.8     | 52.9      | 34.7     | 49.8    | °C/W |

- (1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.
- (2) The package thermal impedance is calculated in accordance with JESD 51-7.

## 6.5 Electrical Characteristics: SNx400

over operating free-air temperature range (unless otherwise noted)

| PARAMETER | TEST CONDITIONS                                                                   | MIN    | TYP | MAX  | UNIT |
|-----------|-----------------------------------------------------------------------------------|--------|-----|------|------|
| $V_{IK}$  | $V_{CC} = \text{MIN}$ and $I_I = -12 \text{ mA}$                                  |        |     | -1.5 | V    |
| $V_{OH}$  | $V_{CC} = \text{MIN}$ , $V_{IL} = 0.8 \text{ V}$ , and $I_{OH} = -0.4 \text{ mA}$ | 2.4    | 3.4 |      | V    |
| $V_{OL}$  | $V_{CC} = \text{MIN}$ , $V_{IH} = 2 \text{ V}$ , and $I_{OL} = 16 \text{ mA}$     |        | 0.2 | 0.4  | V    |
| $I_I$     | $V_{CC} = \text{MAX}$ and $V_I = 5.5 \text{ V}$                                   |        |     | 1    | mA   |
| $I_{IH}$  | $V_{CC} = \text{MAX}$ and $V_I = 2.4 \text{ V}$                                   |        |     | 40   | μA   |
| $I_{IL}$  | $V_{CC} = \text{MAX}$ and $V_I = 0.4 \text{ V}$                                   |        |     | -1.6 | mA   |
| $I_{OS}$  | $V_{CC} = \text{MAX}$                                                             | SN5400 |     | -20  | mA   |
|           |                                                                                   | SN7400 |     | -18  |      |
| $I_{CCH}$ | $V_{CC} = \text{MAX}$ and $V_I = 0 \text{ V}$                                     |        | 4   | 8    | mA   |
| $I_{CCL}$ | $V_{CC} = \text{MAX}$ and $V_I = 4.5 \text{ V}$                                   |        | 12  | 22   | mA   |

## 6.6 Electrical Characteristics: SNx4LS00

over operating free-air temperature range (unless otherwise noted)

| PARAMETER | TEST CONDITIONS                                                                | MIN                                | TYP  | MAX  | UNIT |
|-----------|--------------------------------------------------------------------------------|------------------------------------|------|------|------|
| $V_{IK}$  | $V_{CC} = \text{MIN}$ and $I_I = -18 \text{ mA}$                               |                                    |      | -1.5 | V    |
| $V_{OH}$  | $V_{CC} = \text{MIN}$ , $V_{IL} = \text{MAX}$ , and $I_{OH} = -0.4 \text{ mA}$ | 2.5                                | 3.4  |      | V    |
| $V_{OL}$  | $V_{CC} = \text{MIN}$ and $V_{IH} = 2 \text{ V}$                               | $I_{OL} = 4 \text{ mA}$            | 0.25 | 0.4  | V    |
|           |                                                                                | $I_{OL} = 8 \text{ mA (SN74LS00)}$ | 0.35 | 0.5  |      |
| $I_I$     | $V_{CC} = \text{MAX}$ and $V_I = 7 \text{ V}$                                  |                                    |      | 0.1  | mA   |
| $I_{IH}$  | $V_{CC} = \text{MAX}$ and $V_I = 2.7 \text{ V}$                                |                                    |      | 20   | μA   |
| $I_{IL}$  | $V_{CC} = \text{MAX}$ and $V_I = 0.4 \text{ V}$                                |                                    |      | -0.4 | mA   |
| $I_{OS}$  | $V_{CC} = \text{MAX}$                                                          | -20                                |      | -100 | mA   |
| $I_{CCH}$ | $V_{CC} = \text{MAX}$ and $V_I = 0 \text{ V}$                                  |                                    | 0.8  | 1.6  | mA   |
| $I_{CCL}$ | $V_{CC} = \text{MAX}$ and $V_I = 4.5 \text{ V}$                                |                                    | 2.4  | 4.4  | mA   |

## 6.7 Electrical Characteristics: SNx4S00

over operating free-air temperature range (unless otherwise noted)

| PARAMETER | TEST CONDITIONS                                                                 | MIN | TYP | MAX  | UNIT |
|-----------|---------------------------------------------------------------------------------|-----|-----|------|------|
| $V_{IK}$  | $V_{CC} = \text{MIN}$ and $I_I = -18 \text{ mA}$                                |     |     | -1.2 | V    |
| $V_{OH}$  | $V_{CC} = \text{MIN}$ , $V_{IL} = 0.8 \text{ V}$ , and $I_{OH} = -1 \text{ mA}$ | 2.5 | 3.4 |      | V    |
| $V_{OL}$  | $V_{CC} = \text{MIN}$ , $V_{IH} = 2 \text{ V}$ , and $I_{OL} = 20 \text{ mA}$   |     |     | 0.5  | V    |
| $I_I$     | $V_{CC} = \text{MAX}$ and $V_I = 5.5 \text{ V}$                                 |     |     | 1    | mA   |
| $I_{IH}$  | $V_{CC} = \text{MAX}$ and $V_I = 2.7 \text{ V}$                                 |     |     | 50   | μA   |
| $I_{IL}$  | $V_{CC} = \text{MAX}$ and $V_I = 0.5 \text{ V}$                                 |     |     | -2   | mA   |

## Electrical Characteristics: SNx4S00 (continued)

over operating free-air temperature range (unless otherwise noted)

| PARAMETER | TEST CONDITIONS                                 | MIN | TYP | MAX  | UNIT |
|-----------|-------------------------------------------------|-----|-----|------|------|
| $I_{OS}$  | $V_{CC} = \text{MAX}$                           | –40 |     | –100 | mA   |
| $I_{CCH}$ | $V_{CC} = \text{MAX}$ and $V_I = 0 \text{ V}$   |     | 10  | 16   | mA   |
| $I_{CCL}$ | $V_{CC} = \text{MAX}$ and $V_I = 4.5 \text{ V}$ |     | 20  | 36   | mA   |

## 6.8 Switching Characteristics: SNx400

$V_{CC} = 5 \text{ V}$ ,  $T_A = 25^\circ\text{C}$ , and over operating free-air temperature range (unless otherwise noted). See Figure 2.

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | TEST CONDITIONS                              | MIN | TYP | MAX | UNIT |
|-----------|--------------|-------------|----------------------------------------------|-----|-----|-----|------|
| $t_{PLH}$ | A or B       | Y           | $R_L = 400 \Omega$ and $C_L = 15 \text{ pF}$ |     | 11  | 22  | ns   |
| $t_{PHL}$ |              |             |                                              |     | 7   | 15  |      |

## 6.9 Switching Characteristics: SNx4LS00

$V_{CC} = 5 \text{ V}$ ,  $T_A = 25^\circ\text{C}$ , and over operating free-air temperature range (unless otherwise noted). See Figure 2.

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | TEST CONDITIONS                                     | MIN | TYP | MAX | UNIT |
|-----------|--------------|-------------|-----------------------------------------------------|-----|-----|-----|------|
| $t_{PLH}$ | A or B       | Y           | $R_L = 2 \text{ k}\Omega$ and $C_L = 15 \text{ pF}$ |     | 9   | 15  | ns   |
| $t_{PHL}$ |              |             |                                                     |     | 10  | 15  |      |

## 6.10 Switching Characteristics: SNx4S00

$V_{CC} = 5 \text{ V}$ ,  $T_A = 25^\circ\text{C}$ , and over operating free-air temperature range (unless otherwise noted). See Figure 2.

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | TEST CONDITIONS                              | MIN | TYP | MAX | UNIT |
|-----------|--------------|-------------|----------------------------------------------|-----|-----|-----|------|
| $t_{PLH}$ | A or B       | Y           | $R_L = 280 \Omega$ and $C_L = 15 \text{ pF}$ |     | 3   | 4.5 | ns   |
|           |              |             | $R_L = 280 \Omega$ and $C_L = 50 \text{ pF}$ |     | 4.5 |     |      |
| $t_{PHL}$ | A or B       | Y           | $R_L = 280 \Omega$ and $C_L = 15 \text{ pF}$ |     | 3   | 5   |      |
|           |              |             | $R_L = 280 \Omega$ and $C_L = 50 \text{ pF}$ |     | 5   |     |      |